Linear-algebra verification on the D_IV^5 spin factor — confirms the S2 frame-selection close AND the toy-4900 flaw (frame-invariance ≠ frame-selection), explicitly

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K974 — S2 verified in explicit linear algebra; the toy-4900 correction is airtight

Built the D_IV^5 Euclidean Jordan algebra as the spin factor $V = \mathbb{R}e \oplus \mathbb{R}^4$ with product $(a,v)\circ(b,w) = (ab+\langle v,w\rangle, aw+bv)$, and ran four checks (deterministic, seed 0). All pass.

Results

1. Interior = 2, spectral (rigorous): a generic $x = (a,v)$ has exactly two primitive idempotents $c\pm = ½(e \pm v/|v|)$; verified idempotent ($c\circ c=c$), orthogonal ($c_+\circ c\_=0$), complete ($c_+ + c\_=e$), and $x = \lambda_+ c_+ + \lambda\_c\_$ with $\lambda\pm = a \pm |v|$. Confirms Cal §121 / K973 interior-seat count numerically.
2. ★ The frame is fixed by the operator's DIRECTION (S2 close, FK III.1.2): scaling the element ($x \to 3x+2e$, or $x \to -x$) leaves the frame direction $\hat{u} = v/|v|$ unchanged (up to sign). The direction, not the magnitude, selects the frame.
3. ★ The toy-4900 FLAW confirmed: a central / Jordan-spectral (frame-invariant) symbol ($v=0$) commutes with 5 random DIFFERENT frames — it tests *nothing* about which frame is physical. Frame-invariance (commutes with all) is the opposite of frame-selection. Toy 4900's "$\|[T\_\varphi, \text{frame}]\|=0$ for a color-blind symbol" is the degenerate case, so it does NOT clear S2. (This is the correction from K973, now demonstrated, not asserted.)
4. A NON-CENTRAL operator selects its unique frame: an operator with $v\neq0$ commutes with its OWN frame ($\hat{u}\_op$) and NOT with a transverse frame. So a definite direction picks exactly one frame.

Consequence for S2

The "S^3 continuum" is dissolved *concretely*: the ambiguity exists only for central ($v=0$) elements — a measure-zero locus and exactly the case toy 4900 landed in. S2 = one linear-algebra step: project the F603/K769 condensate direction (SO(5) vector, target-innocent, pinned from quantum numbers) into the spin-factor $\mathbb{R}^4$; if $v \neq 0$, $\hat{u} = v/|v|$ fixes the muon's idempotent frame, target-innocently, and S2 clears. The remaining physics input is F603's direction being non-central in $\mathbb{R}^4$ — a projection, not a continuum.

Audit note: this does NOT bank S2 — it verifies the *mechanism* and corrects the *test*. S2 clears when Lyra/Elie project the actual F603 direction and show $v\neq0$ (non-central). Toy 4900 must be re-run with a non-central condensate symbol, not a frame-invariant one. My K967 verdict stands until then.

Script: run in scratchpad (spin-factor Jordan product + the four checks); Elie to formalize as a claimed toy if wanted.

— K974, Keeper, 2026-07-28. Linear-algebra verification of the S2 close + the toy-4900 correction on the D_IV^5 spin factor. See [[Keeper_K973_BANK_muon_cleared_portions_plus_literature_resolves_both_gates_frame_selection_is_spectra
07-28]], toy 4900.